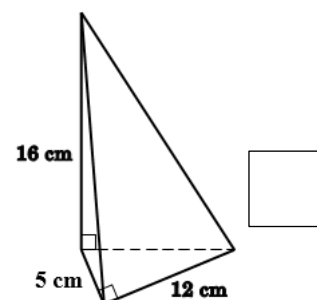


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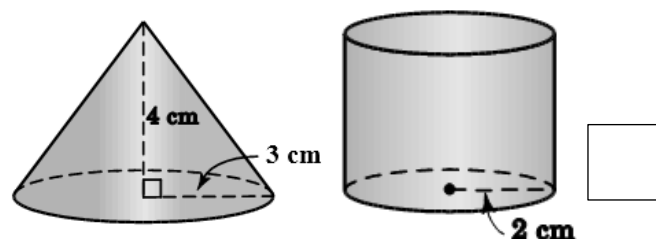
1. What is the volume of pyramid in the figure?

- A. 320 cm^3
B. 230 cm^3
C. 160 cm^3
D. 80 cm^3



2. In the figure, the height of the right circular cone is 4 cm and its base radius is 3 cm. The base radius of the cylinder is 2 cm. If they have the same volume, find the height of the cylinder.

- A. 2 cm
B. 3 cm
C. 4 cm
D. 6 cm



3. 27 small solid metal spheres each of radius 3 cm are melted and recast into a larger solid metal sphere. Find the surface area of the large sphere in terms of π .

- A. $36\pi \text{ cm}^2$
B. $162\pi \text{ cm}^2$
C. $324\pi \text{ cm}^2$
D. $972\pi \text{ cm}^2$



4. The surface area of a sphere is $64\pi \text{ cm}^2$. Find the volume of the sphere.

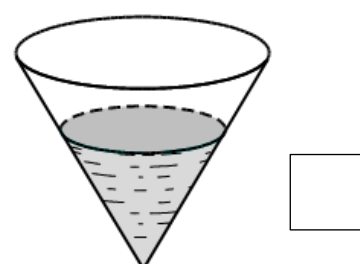
- A. $\frac{64}{3}\pi \text{ cm}^3$
B. $\frac{128}{3}\pi \text{ cm}^3$
C. $64\pi \text{ cm}^3$
D. $\frac{256}{3}\pi \text{ cm}^3$



5. A vessel in the shape of an inverted right circular cone containing 324 cm^3 of water is placed vertically.

If the depth of water in the vessel is $\frac{3}{5}$ of the height of the vessel, find the capacity of the vessel.

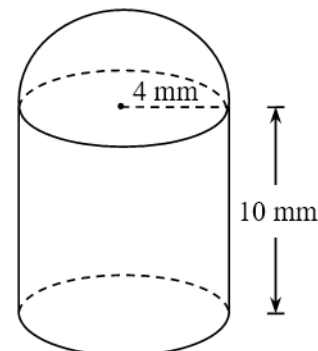
- A. 540 cm^3
B. 750 cm^3
C. 1250 cm^3
D. 1500 cm^3



6. The figure shows a small component of a plastic model. The shapes of its upper part and lower part are a hemisphere and a cylinder respectively. Their base radii are 4 mm and the height of the cylinder is 10 mm. Find the volume of the component.

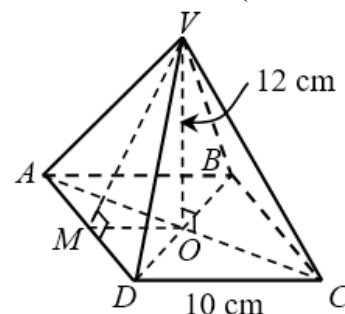
(Give the answer correct to 3 significant figures.)

(3 marks)



7. The height of a solid right pyramid with a square base of side 10 cm is 12 cm. Find the total surface area of the pyramid.

(4 marks)



8. The figure shows a vertical right conical vessel containing some water. The ratio of the curved surface area of the vessel to area of the surface of the vessel in contact with water is $36 : 25$.
- (a) Find the ratio of the base radius of the vessel to the radius of the water surface. (2 marks)
- (b) It is given that the vessel contains 600 cm^3 of water. Alex claims that if 300 cm^3 of water is added into the vessel, the water will overflow. Do you agree? Explain your answer. (5 marks)

